

Submission STAT1-FINAL-SUB01

Question STAT1-FINAL-Q01

scratch: show step

1. $\bar{x} = 26/5 = 5.20$

2. $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 2.20$

=> FINAL: mean=5.20; sample variance=2.20 (check units/interpretation)

Question STAT1-FINAL-Q02

scratch: show step

1. $P(D_1 \cap D_2) = (4/15)((4-1)/(15-1))$

2. =0.0571; the population shrinks after draw 1

=> FINAL: 0.0571 (check units/interpretation)

Question STAT1-FINAL-Q03

scratch: show step

1. $SE = \sqrt{p(1-p)/n} = \sqrt{0.45(1-0.45)/40} = 0.0787$

2. $CI = 0.45 \pm 1.96(0.0787) = (0.296, 0.604)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.296, 0.604); repeated-sampling interpretation (check units/interpretation)

Question STAT1-FINAL-Q04

scratch: show step

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

3. Lower confidence changes coverage, so precision is not free.

=> FINAL: increase n; lower confidence level or variability; note the coverage trade-off (check units/interpretation)

Question STAT1-FINAL-Q05

scratch: show step

1. $0.03 < 0.05$, so reject H_0 under the stated rule.

2. Conclude evidence is inconsistent with H_0 ; do not say the probability H_0 is 3%.

=> FINAL: reject H_0 ; evidence against H_0 , not proof that H_0 is false (check units/interpretation)

Question STAT1-FINAL-Q06

scratch: show step

1. $\bar{x} = 31/5 = 6.20$

2. $s^2 = \sum(x_i - \bar{x})^2 / (5-1) = 1.70$

=> FINAL: mean=6.20; sample variance=1.70 (check units/interpretation)

Question STAT1-FINAL-Q07

scratch: show step

1. $P(D_1 \cap D_2) = (5/19)((5-1)/(19-1))$

2. ≈ 0.0585 ; the population shrinks after draw 1

=> FINAL: 0.0585 (check units/interpretation)

Question STAT1-FINAL-Q08

scratch: show step

1. $SE = \sqrt{p(1-p)/n} = \sqrt{0.35(1-0.35)/80} = 0.0533$

2. $CI = 0.35 \pm 1.96(0.0533) = (0.245, 0.455)$

3. In repeated samples, the procedure captures the population proportion about 95% of the time.

=> FINAL: (0.245, 0.455); repeated-sampling interpretation (check units/interpretation)

Question STAT1-FINAL-Q09

scratch: show step

1. Increase sample size n to reduce standard error.

2. Reduce confidence level or population variability when justified.

3. Lower confidence changes coverage, so precision is not free.

=> FINAL: increase n ; lower confidence level or variability; note the coverage trade-off (check units/interpretation)

Question STAT1-FINAL-Q10

scratch: show step

1. $0.03 < 0.05$, so reject H_0 under the stated rule.

2. Conclude evidence is inconsistent with H_0 ; do not say the probability H_0 is 3%.

=> FINAL: reject H_0 ; evidence against H_0 , not proof that H_0 is false (check units/interpretation)